BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Final Duration: 80 minutes	Semester: Fall 2025 Full Marks: 40

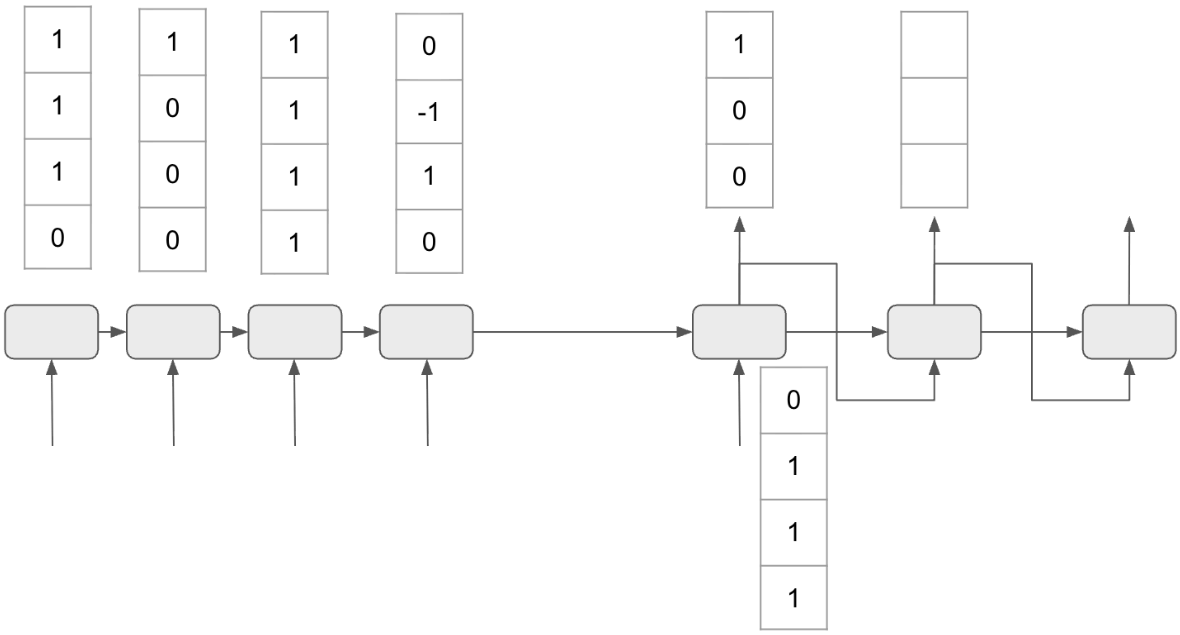
CSE 751: Advanced Natural Language Processing

SET A

Name:	ID:	Section:
Total marks		

You have to answer all of the questions. This question is your answer script– write your answers where indicated. Use the provided answer sheet to do any rough calculations you need to do. Do not write your rough calculations on this sheet.

1. [20]



This is your RNN attention encoder-decoder that uses Bahdanau's additive attention mechanism (which very closely resembles Luong's attention, and is the attention mechanism described in the book). Your job is to generate the output vector for the second (i = 2) hidden unit of the decoder (o_2^d). Here is all the data you have:

$h_1^e = [1\ 1\ 1\ 0]^T$
$h_2^e = [1\ 0\ 0\ 0]^T$
$h_3^e = [1\ 1\ 1\ 1]^T$
$h_4^e = [0\ {-1}\ 1\ 0]^T$
$h_1^d = [0\ 1\ 1\ 1]^T$
$W^d = [0\ 0\ 0,\ 1\ 0\ 0,\ 1\ 1\ 1,\ 1\ 0\ 1]$
$U^d = [1\ 1\ 1\ 1,\ 0\ 0\ 0\ 0,\ 1\ 0\ 1\ 0,\ 0\ 1\ 0\ 1]$
$V^d = [1\ 1\ 0\ 0,\ 0\ 0\ 0\ 0,\ 1\ 1\ 1\ 0]$
$o_1^d = [1\ 0\ 0]^T = \hat{y}_1$

$h_i^d = \tanh(W^d\hat{y}_{i-1} + Uh_{i-1}^d + c_i)$
$\alpha_{ij} = \text{softmax}(\text{score}(h_{i-1}^d, h_j^e))$ for all i and j
$\text{score}(h_i^d, h_j^e) = h_i^d . h_j^e$ for all i and j
$c_i = \sum\alpha_{ij}h_j^e$ for all i and j
$o_i^d = \text{softmax}(V^d h_i^d)$

Fill in the empty vector in the figure above. Show your work on the next page.

2. Each answer is worth 5. Use this page only. [10]
   a. How do you do float16 and int8 quantization from float32?
   b. What is retrieval augmented generation (RAG)? What are the use cases for RAG?

3. Each question is worth 5. Use this page only. [10]
    a. How can we do byte pair encoding tokenization? Explain with an example.
    b. Why do we need GPUs to train transformer models?